

# Fine-tuning NLLB for Spanish-to-Asturian Machine Translation with different data sizes and validation methods

Iker Gutierrez Fandiño

University of the Basque Country (EHU)

igutierrez134@ikasle.ehu.eus

## 1 Introduction

Asturian is a Romance language spoken in the Asturias region of northern Spain, with approximately 100,000 speakers. Despite its linguistic proximity to Spanish, it remains severely under-resourced in the NLP landscape, with very limited parallel corpora and few dedicated machine translation (MT) systems.

This work addresses the task of Spanish-to-Asturian machine translation by fine-tuning NLLB-200-distilled-600M (NLLB Team et al., 2022), a massively multilingual sequence-to-sequence model released by Meta AI. The pipeline was developed in two stages: a prototyping phase on Google Colab with subsampled data, followed by full-scale training on the HiTZ center’s AZTI server (42GB of GPU memory). All the code is available on GitHub and the fine-tuned models are available on HuggingFace (see Appendix A for more details).

## 2 Pipeline

Figure 1 illustrates the overall pipeline. After data collection and tokenization, the base model is used directly for baseline inference, while also serving as the starting point for LoRA fine-tuning (Hu et al., 2022) with three alternative validation strategies (validation loss, BLEU, and chrF++). Both the baseline and fine-tuned models are then evaluated automatically using BLEU, chrF++, COMET, and BLEURT.

### 2.1 Data

The training data was sourced from the ES-AST Parallel Corpus from [Proyecto AINA \(2023\)](#). For development and evaluation, the FLORES+ benchmark ([Guzmán et al., 2021](#)) was used, providing standardized dev and devtest splits for both Spanish (`spa_Latn`) and Asturian (`ast_Latn`).

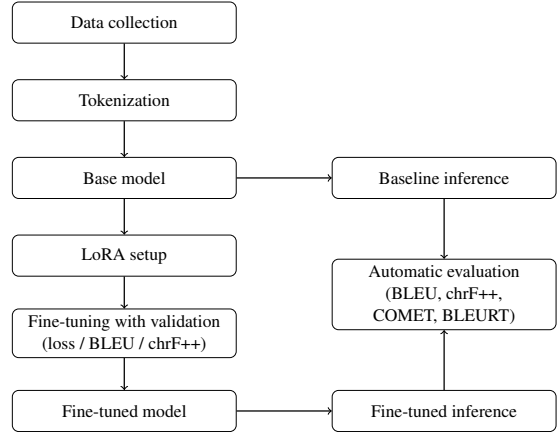


Figure 1: Overview of the MT fine-tuning pipeline.

In the Colab prototyping phase, the training set was subsampled to 5,000 sentence pairs, while the development and test sets were capped at 500 examples each, to allow rapid iteration within the Colab resource constraints. In the server experiments, the full datasets were used for training, development and evaluation, as shown in Table 1:

| Split | Corpus  | Colab | Server  |
|-------|---------|-------|---------|
| Train | AINA    | 5,000 | 704,378 |
| Dev   | FLORES+ | 500   | 997     |
| Test  | FLORES+ | 500   | 1,012   |

Table 1: Number of sentence pairs per split in the Colab and server experimental settings.

All data was tokenized using the NLLB tokenizer with `spa_Latn` as the source language, with a maximum sequence length of 96 tokens in Colab and 128 tokens on the server. A `forced_bos_token_id` corresponding to `ast_Latn` was set on the model’s generation configuration to constrain decoding to Asturian.

## 2.2 Model and Parameter-Efficient Fine-Tuning

The base model is *facebook/nllb-200-distilled-600M*, a distilled variant of NLLB-200 with 600M parameters. Since full fine-tuning of such a model is computationally expensive, LoRA was applied to the attention projection layers (`q_proj`, `k_proj`, `v_proj`, `out_proj`), thus reducing the number of trainable parameters to a small fraction of the total while preserving the pretrained representations.

In the Colab phase, LoRA was configured with rank  $r = 8$  and scaling factor  $\alpha = 16$ . On the server, these were increased to  $r = 16$  and  $\alpha = 32$  to provide greater representational capacity given the larger training set. The model was loaded in `bfloat16` precision on the server (supported by the GPU capacity) and in `float16` on Colab.

## 2.3 Training

**Colab experiments.** Training was conducted using a batch size of 8 per device with no gradient accumulation, a learning rate of  $5 \times 10^{-5}$ , and 1.5 epochs. Validation was performed every 100 steps using loss only (`predict_with_generate=False`), with no metric-based model selection. This setup allowed rapid prototyping at low computational cost.

**Server experiments.** Three separate training runs were conducted on the server, each differing in the validation criterion used for model selection:

- **Loss validation:** model checkpoint selected by lowest validation loss.
- **BLEU validation:** checkpoint selected by highest corpus BLEU score on the development set.
- **chrF++ validation:** checkpoint selected by highest SacreBLEU chrF++ score (`word_order=2`) on the development set.

All server runs used a batch size of 16 per device, gradient accumulation over 2 steps (effective batch size 32), a cosine learning rate schedule with  $2 \times 10^{-4}$  peak learning rate, 5% warmup ratio, weight decay of 0.01, and 1.5 training epochs. Evaluation was performed every 2,000 steps with beam search decoding (4 beams) during validation. Gradient checkpointing with `use_reentrant=False` was enabled to reduce memory usage.

## 2.4 Inference

Inference was performed in batches of 32 on the server GPU. The fine-tuned LoRA adapter was loaded on top of the base model using `PeftModel.from_pretrained`. Generation used 5 beams, a repetition penalty of 1.15, a trigram no-repeat constraint, and the `forced_bos_token_id` was set to `ast_Latn`.

## 2.5 Evaluation

All models were evaluated on the FLORES+ devtest split using SacreBLEU’s BLEU and chrF++, along with COMET and BLEURT as neural evaluation metrics. Table 2 reports the results:

| Model                      | BLEU         | chrF++       | COMET        | BLEURT       |
|----------------------------|--------------|--------------|--------------|--------------|
| <i>Colab (subsampling)</i> |              |              |              |              |
| Baseline (NLLB)            | 15.32        | 42.17        | –            | –            |
| FT (loss val)              | 12.80        | 37.45        | –            | –            |
| <i>Server (full data)</i>  |              |              |              |              |
| Baseline (NLLB)            | 13.88        | 40.80        | 65.97        | 44.71        |
| FT (loss val)              | <b>16.36</b> | 45.80        | 67.28        | 43.34        |
| <b>FT (BLEU val)</b>       | 16.32        | <b>46.09</b> | <b>68.27</b> | <b>44.98</b> |
| FT (chrF++ val)            | 16.26        | 45.48        | 66.95        | 42.70        |

Table 2: Evaluation results on FLORES+ devtest (SPA→AST).

Results show three main findings. First, with scarce data (5,000 sentences), fine-tuning hurts performance: the Colab baseline outperforms the fine-tuned model across all metrics. Second, with full data, fine-tuning consistently improves over the baseline. Third, the choice of validation strategy has little impact for this language pair: BLEU validation performs best, but loss and chrF++ validation are close behind.

## 3 Conclusion

This work has presented a parameter-efficient approach for Spanish-to-Asturian machine translation by fine-tuning NLLB-200 with LoRA. Results have shown that fine-tuning on scarce data drops performance, whereas fine-tuning on the full corpus consistently improves over the baseline. The choice of validation strategy has little impact, with BLEU validation performing slightly best. Future work could explore larger LoRA ranks to increase model capacity, or additional parallel data to further improve translation quality.

## References

- Francisco Guzmán, Peng-Jen Chen, Myle Ott, Juan Pino, Guillaume Lample, Philipp Koehn, Vishrav Chaudhary, and Marc’Aurelio Ranzato. 2021. [FLORES+: A Benchmark Dataset for Machine Translation in Low-Resource Languages](#). *Machine Translation*, 35(1):61–90.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. *arXiv preprint arXiv:2106.09685*.
- NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Heffernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, Anna Sun, Skyler Wang, Guillaume Wenzek, Al Youngblood, Bapi Akula, Loïc Barrault, Gabriel Mejia-Gonzalez, Prangthip Hansanti, John Hoffman, Semarley Jarrett, Kaushik Ram Sadagopan, Dirk Rowe, Shannon Spruit, Chau Tran, Pierre Andrews, Necip Fazil Ayan, Shruti Bhosale, Sergey Edunov, Angela Fan, Cynthia Gao, Vedanuj Goswami, Francisco de Guzman, Philipp Koehn, Alexandre Mourachko, Christophe Ropers, Safiyyah Saleem, Holger Schwenk, and Jeff Wang. 2022. [No Language Left Behind: Scaling Human-Centered Machine Translation](#). *arXiv preprint arXiv:2207.04672*.
- Projecte AINA. 2023. ES-AST Parallel Corpus. [https://huggingface.co/datasets/projecte-aina/ES-AST\\_Parallel\\_Corpus](https://huggingface.co/datasets/projecte-aina/ES-AST_Parallel_Corpus).

## A Resources

All code and models produced in this work are publicly available at the following links:

- **GitHub repository** (scripts, predictions, results):  
<https://github.com/iker-gutierrez/es-ast-MT>
- **LoRA adapter for BLEU validation** (HuggingFace Hub):  
<https://huggingface.co/ikergf/asturian-nllb-lora-bleu>
- **LoRA adapter for loss validation** (HuggingFace Hub):  
<https://huggingface.co/ikergf/asturian-nllb-lora-loss>
- **LoRA adapter for chrF++ validation** (HuggingFace Hub):  
<https://huggingface.co/ikergf/asturian-nllb-lora-chrfpp>